

Tea Blending Optimisation: A Comprehensive Analysis

STA5074Z Prescriptive Analytics

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1 Introduction

Blending problems are a classic challenge in operations research, arising in diverse fields such as food processing, metallurgy, and energy. In the tea industry, blending is crucial to achieve consistent product quality, balance raw material costs, and adapt to fluctuating supplies. Every cup of tea on the supermarket shelf is the result of a carefully orchestrated process. Different grades and types of tea leaves from various origins are combined to meet precise targets for taste, colour, aroma and other quality parameters, all while minimising production costs and ensuring supply constraints are not violated.

In the context of tea manufacturing, the blending optimisation problem involves selecting the optimal composition of raw teas to satisfy desired blend qualities for multiple final blends. This not only impacts the sensory profile and brand identity of each product, but also has substantial operational and economic implications. By applying advanced computational techniques, such as mathematical programming and metaheuristics, producers can automate and enhance the blending decision-making process, yielding better margins, consistent quality and greater responsiveness to market demands.

Below, we formally define the tea blending optimisation problem and systematically explore several approaches for solving it. The problem requires determining the least-cost combination of raw materials to produce final blends that meet specific quality characteristics, as formalised by Fomeni^[1].

The primary objectives are:

1. To implement and solve a cost-minimisation model using Linear Programming (LP).
2. To solve the same problem using Simulated Annealing (SA) metaheuristic and compare performance.
3. To formulate and solve a multi-objective model using Chebyshev Goal Programming.
4. To explore Genetic Algorithms (GAs) as an alternative solution approach.

2 Problem Formulation and Data Preparation

2.1 Notations and Parameters

- **Indices:**
 - i : Index for raw materials
 - j : Index for final blends
 - k : Index for quality characteristics
- **Parameters:**
 - P_i : Unit cost of raw material i (£/Kg)
 - D_j : Weekly demand for final blend j (Kgs)
 - a_i : Weekly availability of raw material i (Kgs)
 - g_{ik} : Grade of characteristic k in one unit of raw material i
 - s_{jk} : Required target score of characteristic k in one unit of final blend j
- **Decision Variables:**
 - x_{ij} : Amount (in Kgs) of raw material i to be used in final blend j

2.2 Data Source

This project uses the real-world data from Fomeni^[1], extracted from tables 2 and 3 of the paper. These data represent actual tea blending scenarios from a UK-based tea company. The data are loaded and prepared in

section 3 below.

3 Data Implementation (Fomeni 2018)

This section implements the tea blending optimisation problem using the actual data provided in Fomeni^[1]. The data consist of 60 raw materials and 10 final blends, each with 6 quality characteristics.

3.1 Data Loading

The dataset comprises **60 raw materials** and **10 final blends**, each characterised by **6 quality attributes** (C1 through C6). Raw material prices range from £0.0120 to £0.0470 per kilogram, with a mean price of £0.0217 per kilogram. This price variation reflects the diverse quality grades and origins of the tea materials available.

Table 1: Raw Materials: Costs (£/kg), Availability (kg), and Quality Characteristics (First 10)

Material_ID	Cost	Availability	C1	C2	C3	C4	C5	C6
R1	0.0240	5,000	5	4	9	6	5	5
R2	0.0167	1,000	5	3	9	4	5	1
R3	0.0235	25,000	5	4	9	6	5	1
R4	0.0170	1,000	5	3	9	4	5	1
R5	0.0230	25,000	5	4	9	6	5	1
R6	0.0265	20,000	7	4	9	6	7	1
R7	0.0265	20,000	7	4	9	6	7	1
R8	0.0294	1,000	8	4	9	6	8	5
R9	0.0284	15,000	8	4	9	6	8	1
R10	0.0284	15,000	8	4	9	6	8	1

Table 2: Blends: Demand (kg) and Target Quality Scores

Blend_ID	Demand	C1	C2	C3	C4	C5	C6
B1	100,000	2.3	3.0	3.0	2.0	2.32	1
B2	100,000	3.8	4.7	2.9	3.0	3.80	1
B3	60,000	6.6	5.2	6.1	6.9	6.60	1
B4	100,000	6.2	5.2	5.3	6.4	6.20	1
B5	100,000	4.7	5.1	4.5	5.0	4.70	1
B6	50,000	8.4	4.7	6.7	7.3	8.40	1
B7	10,000	4.0	0.5	1.0	5.0	4.00	1
B8	5,000	5.9	0.5	1.0	3.9	5.90	1
B9	10,000	7.0	3.9	9.0	5.9	7.00	1

Table 2: Blends: Demand (kg) and Target Quality Scores

Blend_ID	Demand	C1	C2	C3	C4	C5	C6
B10	20,000	4.0	2.0	3.0	2.0	4.00	9

A feasibility analysis reveals that the total raw material availability is **830,500 kg**, while total demand across all blends is **555,000 kg**. This represents a **49.64% surplus** in availability, indicating that supply constraints are not binding and the problem has sufficient flexibility for optimisation. The surplus availability provides the solver with multiple feasible solution paths, which is advantageous for finding cost-optimal allocations.

3.2 Data Characteristics

The following analysis provides a comprehensive overview of the dataset's key characteristics, enabling a deeper understanding of the problem structure before optimisation.

Table 3: Price Distribution Statistics

Statistic	Value
Mean	£0.0217
Median	£0.0195
Minimum	£0.0120
Maximum	£0.0470
Standard Deviation	£0.0076

Table 4: Availability Distribution Statistics (kg)

Statistic	Value_kg
Mean	13842
Median	5000
Minimum	500
Maximum	80000
Total	830500

Table 5: Quality Characteristics Summary Statistics

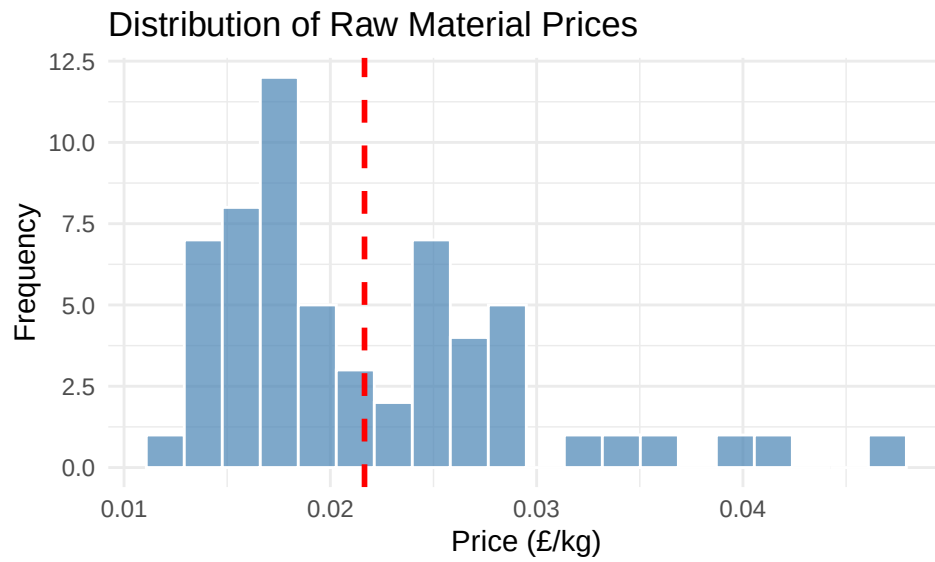
Characteristic	Minimum	Maximum	Mean	Median	Std_Dev
C1	2.0	9	5.500000	5.0	1.917802
C2	0.5	8	3.333333	3.0	1.990798
C3	1.0	9	4.316667	4.0	2.507559
C4	2.0	8	4.500000	3.5	2.452947

Table 5: Quality Characteristics Summary Statistics

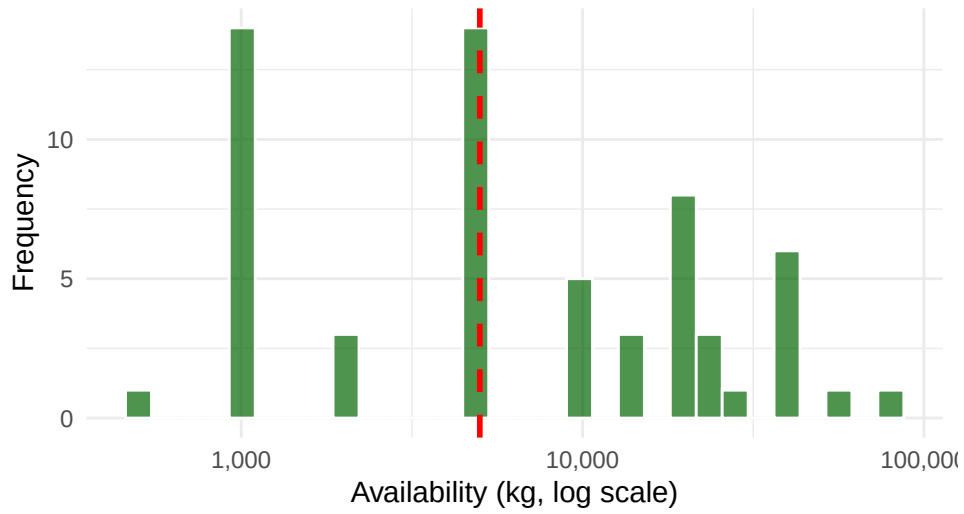
Characteristic	Minimum	Maximum	Mean	Median	Std_Dev
C5	2.0	9	5.500000	5.0	1.917802
C6	1.0	9	1.933333	1.0	1.858254

Table 6: Blend Demand Statistics (kg)

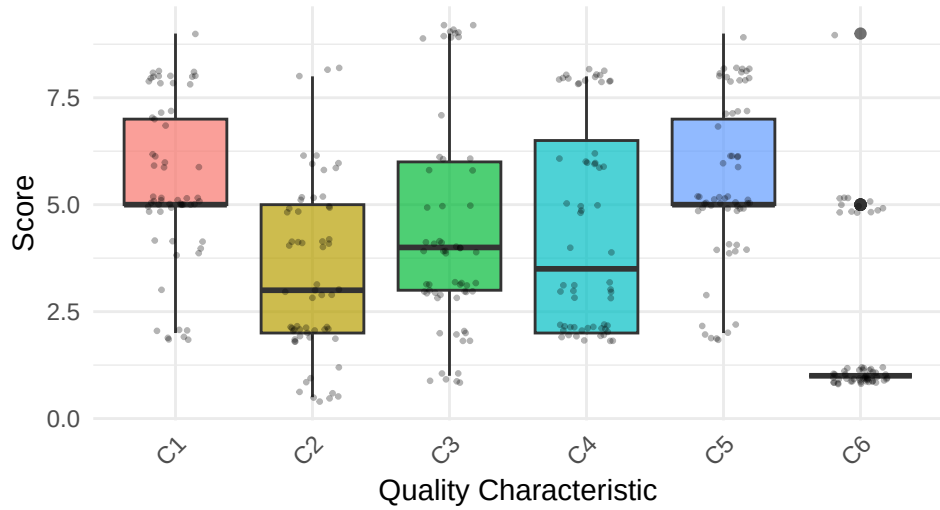
Statistic	Value_kg
Mean	55500
Median	55000
Minimum	5000
Maximum	100000
Total	555000



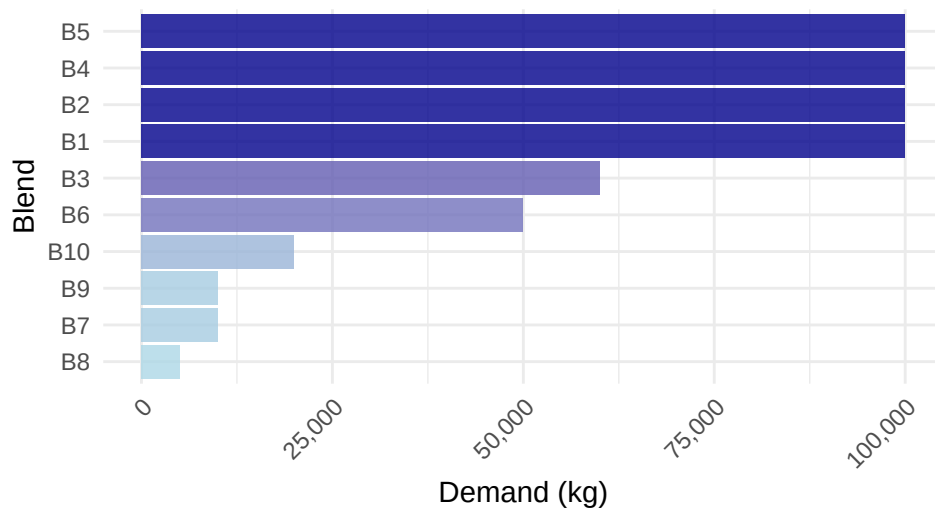
Distribution of Raw Material Availability

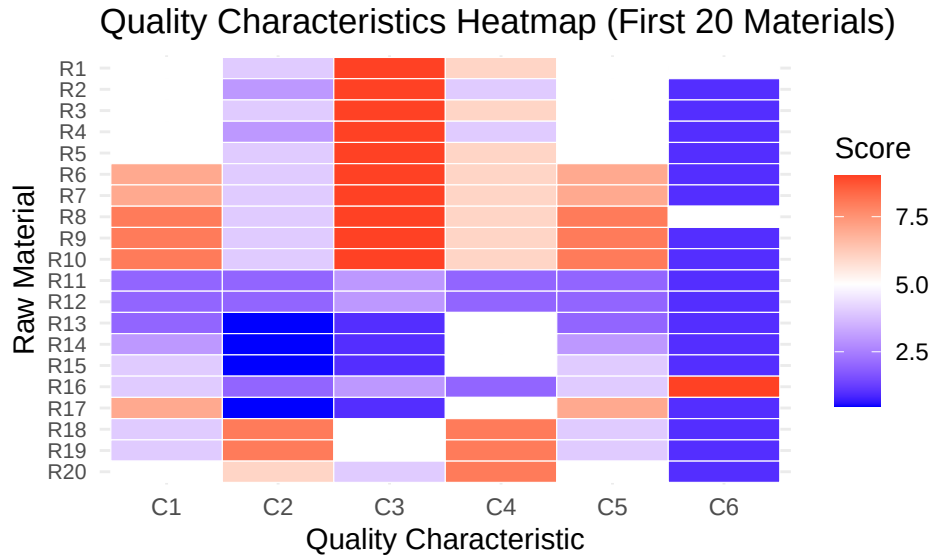
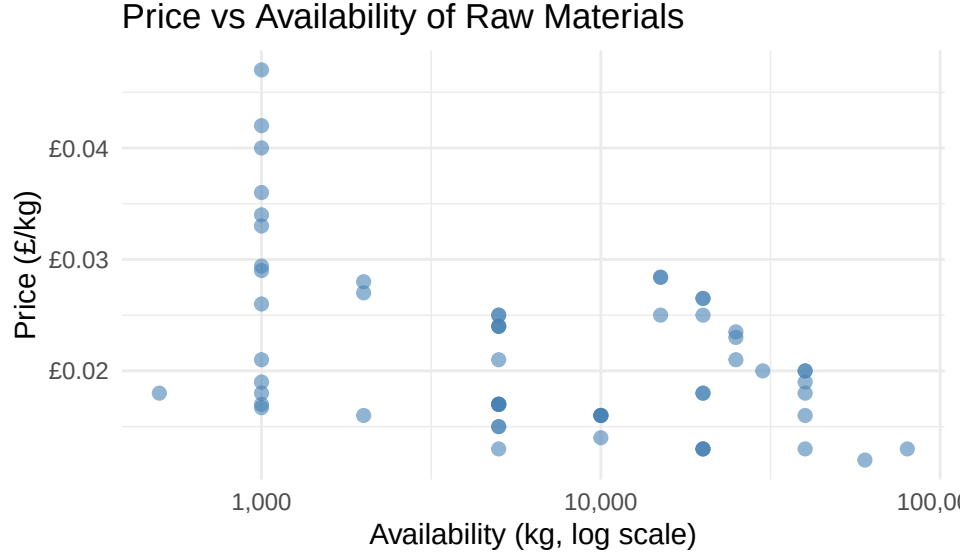


Distribution of Quality Characteristics Across Raw Material



Demand by Blend





4 Approach 1: Cost Minimisation using Linear Programming

This section implements the cost-minimisation model using Linear Programming with parametric relaxation of the characteristic-matching constraints. This involves implementing a cost-minimisation LP model that finds the optimal combination of raw materials to produce blends meeting demand and quality constraints. This implementation uses Rglpk which provides a straightforward matrix-based formulation that is well-suited for this problem type.

4.1 Model Formulation

Objective Function: Minimise total cost

$$\min \sum_i \sum_j P_i x_{ij}$$

Subject to Constraints:

1. Raw Material Availability: $\sum_j x_{ij} \leq a_i \quad \forall i$

2. Blend Demand: $\sum_i x_{ij} \geq D_j \quad \forall j$
3. Characteristic Score Relaxation:
 - $\sum_i (g_{ik} - (s_{jk} + \epsilon_{jk}))x_{ij} \leq 0 \quad \forall j, k$
 - $\sum_i (g_{ik} - (s_{jk} - \epsilon_{jk}))x_{ij} \geq 0 \quad \forall j, k$
4. Non-negativity: $x_{ij} \geq 0 \quad \forall i, j$

4.2 Implementation

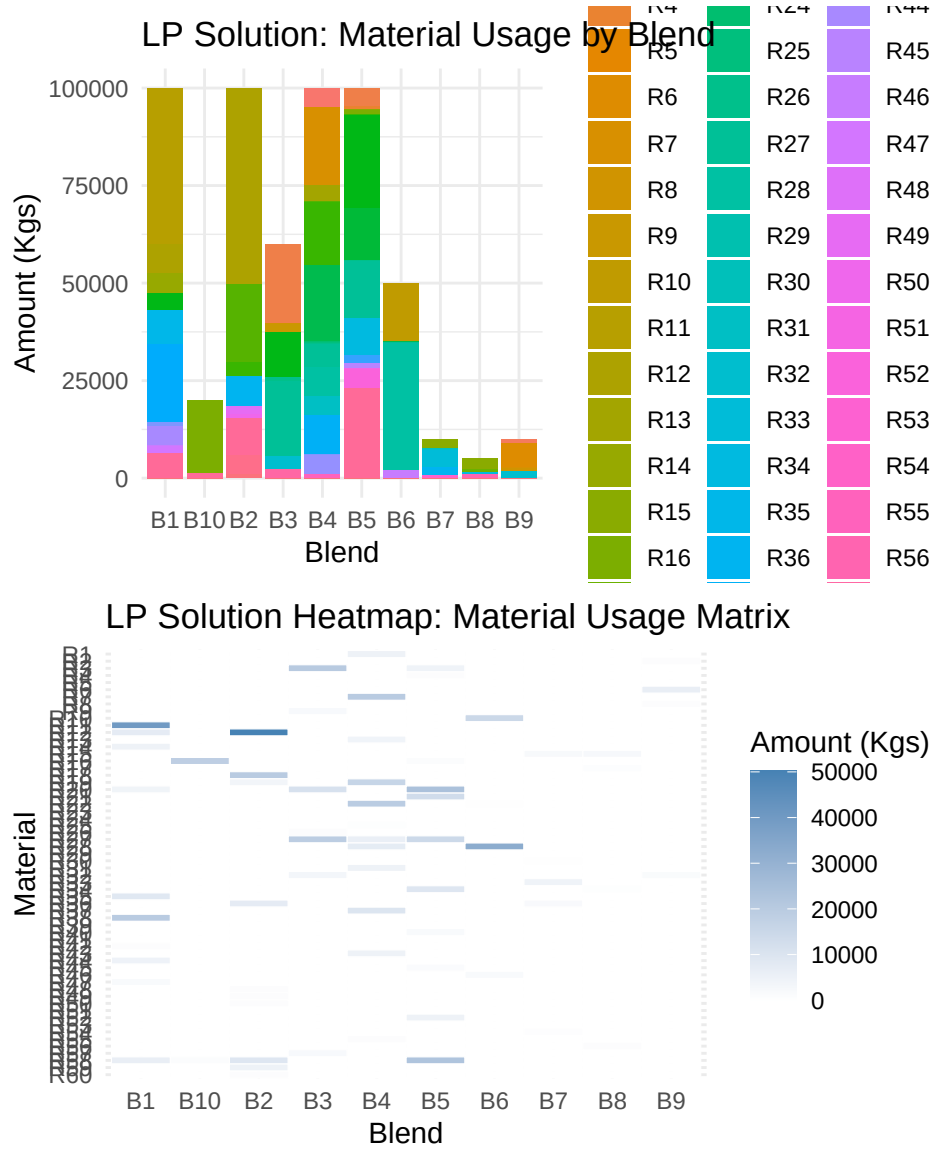
The Linear Programming model was successfully solved using the Rglpk solver, which employs the GNU Linear Programming Kit (GLPK) algorithm. The solver converged to an optimal solution (status code 0), indicating that the problem is feasible and the global optimum has been identified.

Table 7: Absolute deviations from target quality scores

Blend	C1	C2	C3	C4	C5	C6
B1	0.5000000	0.5000000	0.05429705	0.5000000	0.4800000	0.19873016
B2	0.5000000	0.5000000	0.5000000	0.5000000	0.5000000	0.12126984
B3	0.3736758	0.5000000	0.5000000	0.1152129	0.3736758	0.06666667
B4	0.5000000	0.43809524	0.24592593	0.1150794	0.5000000	0.2400000
B5	0.5000000	0.32182063	0.5000000	0.5000000	0.5000000	0.1000000
B6	0.5000000	0.07333333	0.10666667	0.1400000	0.5000000	0.1600000
B7	0.3977401	0.5000000	0.5000000	0.5000000	0.3977401	0.23728814
B8	0.5000000	0.2500000	0.5000000	0.3000000	0.5000000	0.0000000
B9	0.2333333	0.16666667	0.5000000	0.2333333	0.2333333	0.4000000
B10	0.0000000	0.3750000	0.0000000	0.0000000	0.0000000	0.5000000

The Linear Programming solution achieves an **optimal total cost of £9,087.69**, utilising exactly **555,000 kg** of raw materials to meet the total demand requirement. The solution employs **52 out of 60 available raw materials**, demonstrating that the optimal strategy does not require the full material portfolio. The materials utilised in the optimal solution include R1, R2, R3, R4, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R25, R26, R27, R28, R30, R31, R32, R33, R34, R35, R36, R37, R38, R40, R42, R43, R44, R45, R46, R47, R48, R49, R50, R52, R54, R55, R56, R57, R58, R59, and R60. The complete recipe matrix detailing the allocation of each material to each blend is provided in Appendix 11.1.

4.3 Visualisation



5 Approach 2: Cost Minimisation using Simulated Annealing

Simulated Annealing is a metaheuristic that explores the solution space probabilistically, accepting worse solutions with decreasing probability as the algorithm “cools down.”

5.1 Implementation

Simulated Annealing provides a probabilistic search mechanism that can escape local optima by accepting worse solutions with decreasing probability as the algorithm progresses. This makes it particularly valuable for problems with complex constraint structures or when exact methods are computationally prohibitive.

Table 8: Absolute deviations from target quality scores

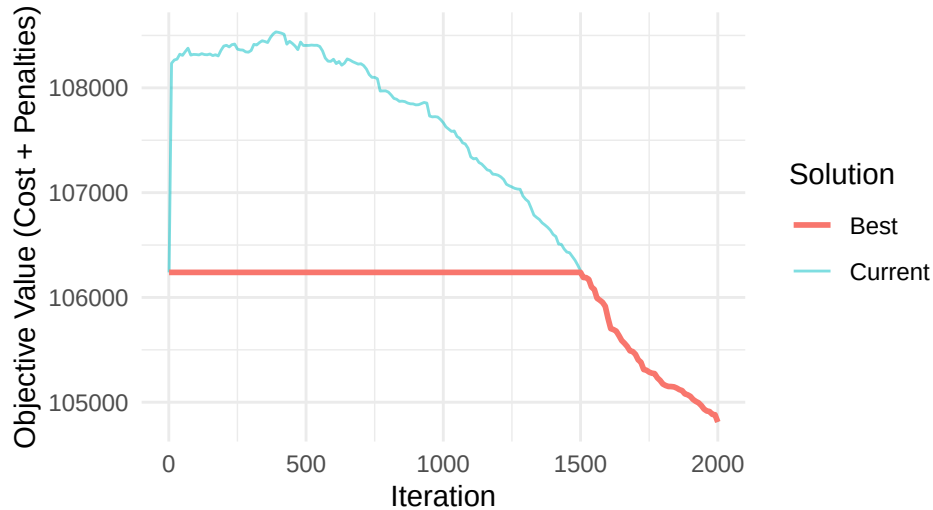
Blend	C1	C2	C3	C4	C5	C6
B1	2.9764445	0.3815626	1.6228325	3.1139699	2.9564445	0.4715045
B2	1.9225550	0.9155676	2.0293568	2.6340229	1.9225550	0.1980452
B3	1.4202349	1.4516624	1.5652475	1.6184567	1.4202349	0.4109966
B4	1.2172613	1.0479202	0.7513509	1.1983750	1.2172613	0.5423994
B5	0.5249761	1.0502650	0.3617898	0.2641280	0.5249761	0.5588779
B6	3.2842225	0.8250129	2.1349574	2.3310344	3.2842225	0.3567670
B7	1.6296551	3.5684759	3.9741594	0.2245883	1.6296551	0.4025894
B8	0.5447635	3.2769350	4.1991170	1.4592010	0.5447635	0.2300009
B9	1.6182375	0.1370972	4.1303345	0.9482161	1.6182375	0.3960921
B10	1.4239205	1.6547675	1.3783382	3.1491767	1.4239205	7.7670997

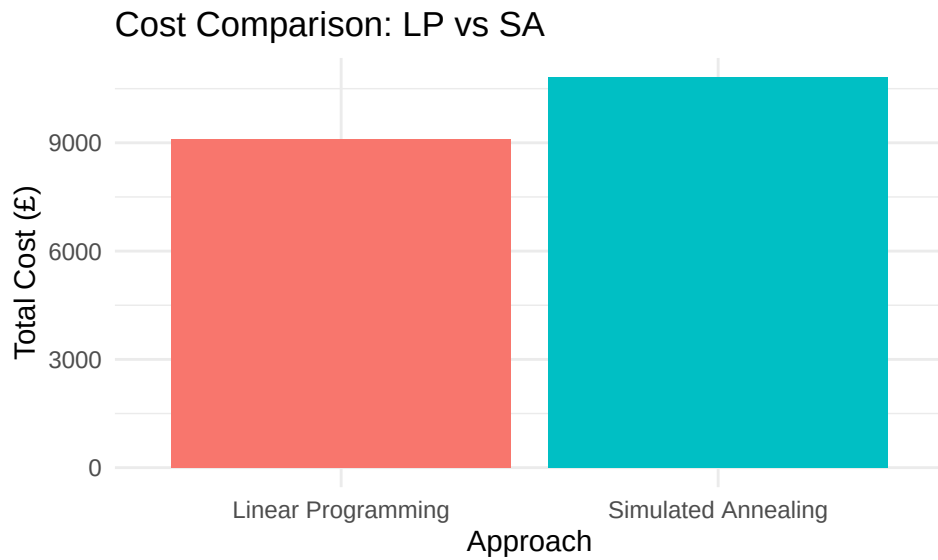
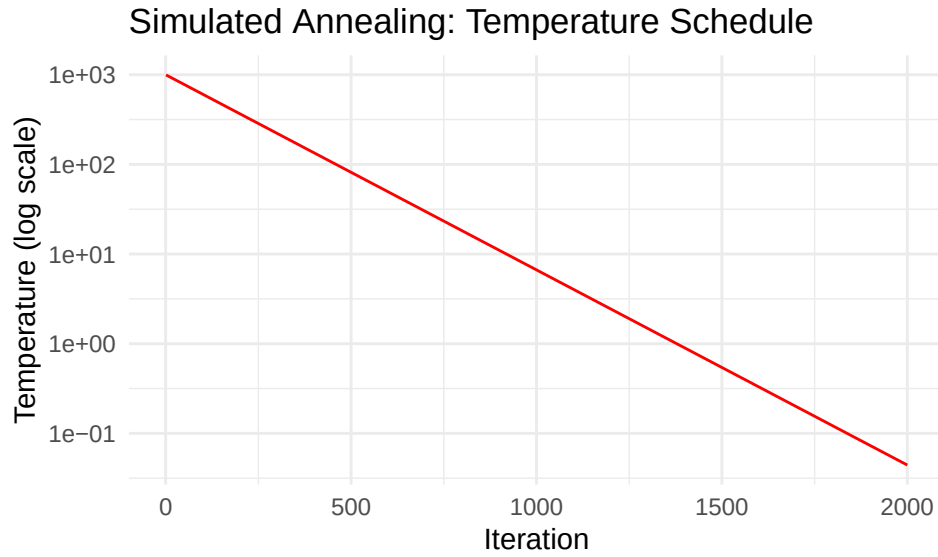
Simulated Annealing was executed with an initial temperature of 1000, a cooling rate of 0.995, and a maximum of 2000 iterations. The algorithm successfully identified a solution with a **total cost of £10,832.36**, representing a **19.20% increase** compared to the Linear Programming optimal solution. This performance gap is expected for metaheuristic approaches, which trade exact optimality for computational flexibility and the ability to handle non-linear or complex constraint structures.

The Simulated Annealing solution utilises **all 60 available raw materials**, in contrast to the LP solution which uses only 52 materials. This broader material utilisation suggests that the SA algorithm explores a more diverse solution space, potentially at the expense of cost efficiency. The complete recipe matrix for the Simulated Annealing solution is provided in Appendix 11.2.

5.2 Visualisation

Simulated Annealing: Convergence Plot





6 Approach 3: Multi-objective Optimisation using Chebyshev Goal Programming

Chebyshev Goal Programming minimises the worst-case deviation from multiple goals, providing a balanced solution that trades off cost for quality control.

6.1 Model Formulation

Objective Function: Minimise worst-case deviation

$$\min \max\{w_0 \cdot d_0, w_{jk} \cdot d_{jk}\}$$

Where: - d_0 : Deviation from cost goal - d_{jk} : Deviation from quality goal for blend j , characteristic k

Constraints: - All LP constraints - Goal constraints for cost and quality

6.2 Implementation

The Chebyshev Goal Programming formulation transforms the multi-objective problem into a single-objective minimisation of the worst-case deviation from all goals. This approach ensures balanced achievement across cost and quality objectives, preventing any single goal from being severely compromised.

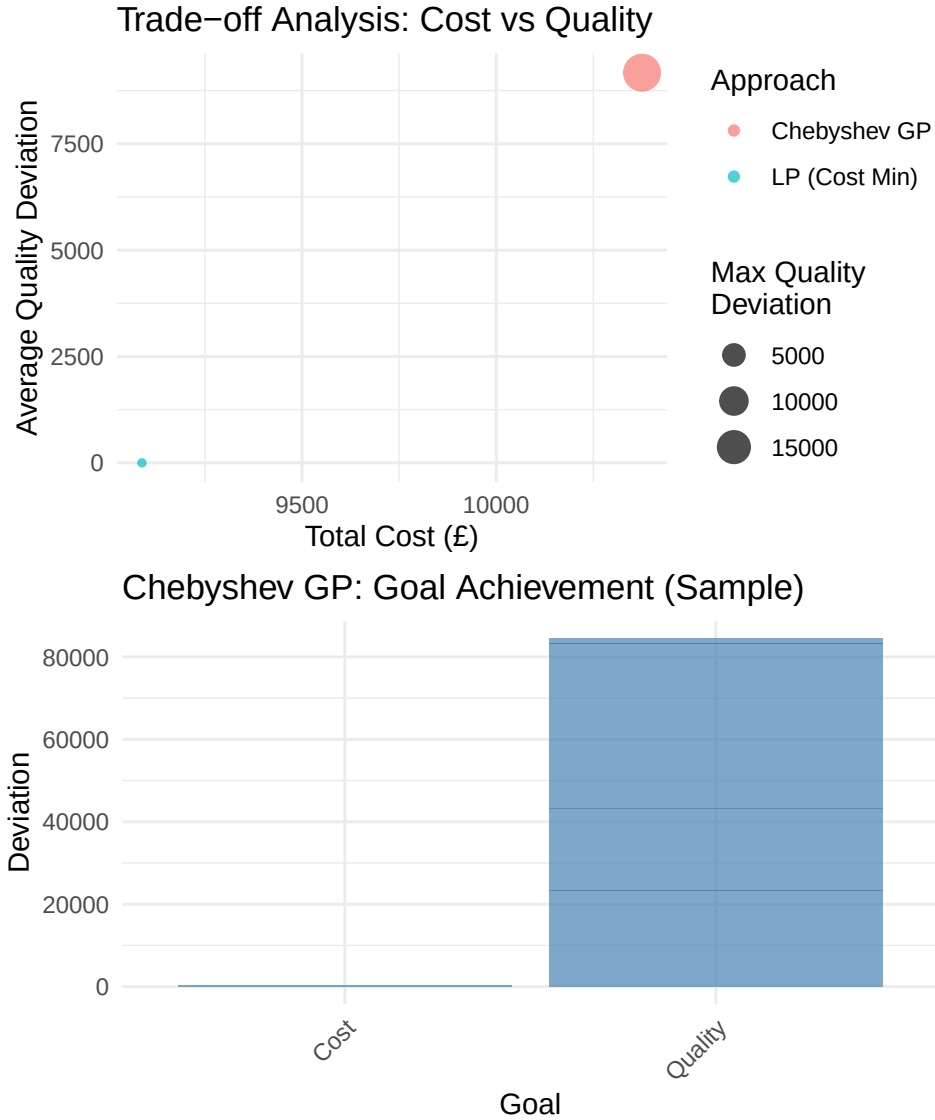
Table 9: Quality deviations (minimised in worst-case sense)

Blend	C1	C2	C3	C4	C5	C6
B1	3,324.17181172407435952	19,972.515	6,657.505	19,972.515	5,324.17181172407345002	0.00
B2	19,972.51543517221944057	19,972.515	19,972.515	19,972.515	19,972.51543517226673430	19,972.52
B3	0.00000000002763036	19,972.515	0.000	16,297.229	0.00000000002763036	0.00
B4	19,972.51543517226309632	19,972.515	19,972.515	19,972.515	19,972.51543517226673430	19,972.52
B5	19,972.51543517226673430	19,972.515	19,972.515	19,972.515	19,972.51543517226673430	19,972.52
B6	19,972.51543517226309632	1,761.103	16,742.083	19,972.515	19,972.51543517226673430	4,000.00
B7	0.00000000000000000	0.000	0.000	0.000	0.00000000000000000	0.00
B8	0.00000000000000000	0.000	0.000	5,500.000	0.00000000000000000	0.00
B9	1,250.00000000000000000	500.000	1,000.000	0.000	1,250.00000000000000000	0.00
B10	0.00000000000000000	0.000	0.000	821.115	0.00000000000000000	6,568.92

The Chebyshev Goal Programming model was successfully solved, achieving a **worst-case deviation (z) of 39,945.03**. The solution yields a **total cost of £10,375.59**, representing a **14.17% increase** over the Linear Programming optimal cost. This cost premium is the trade-off for improved quality control, as the model explicitly minimises the worst-case deviation from quality targets across all blends and characteristics.

The cost deviation from the target (set at 110% of LP optimal cost) is **£379.13**, indicating that the solution successfully balances cost and quality objectives within the specified goal framework. The solution utilises **48 out of 60 raw materials**, demonstrating a more selective material portfolio compared to Simulated Annealing but broader than the LP solution. The materials used include R2, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R27, R29, R32, R33, R35, R36, R38, R39, R40, R42, R43, R44, R45, R46, R47, R48, R49, R50, R51, R52, R54, R55, R56, R58, R59, and R60. The complete recipe matrix is provided in Appendix 11.3.

6.3 Visualisation



7 Approach 4: Genetic Algorithms

Genetic Algorithms use evolutionary principles to evolve a population of solutions toward better fitness.

7.1 Implementation

Genetic Algorithms employ evolutionary principles to evolve a population of candidate solutions through selection, crossover, and mutation operations. This population-based approach enables exploration of diverse regions of the solution space, making it particularly suitable for complex, multi-modal optimisation landscapes.

Table 10: Absolute deviations from target quality scores

Blend	C1	C2	C3	C4	C5	C6
B1	2.466458904	0.7118394	1.3488140	3.097280878	2.446458904	0.2754976

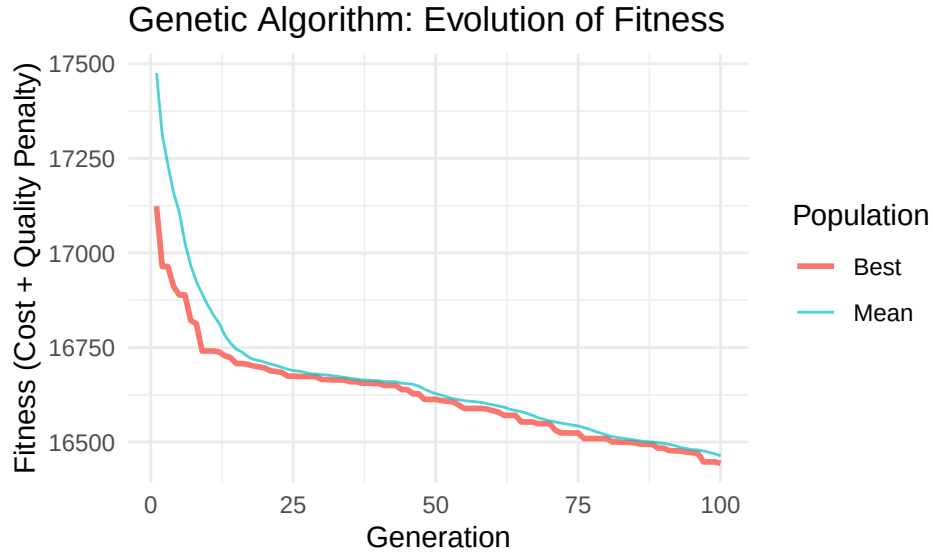
Table 10: Absolute deviations from target quality scores

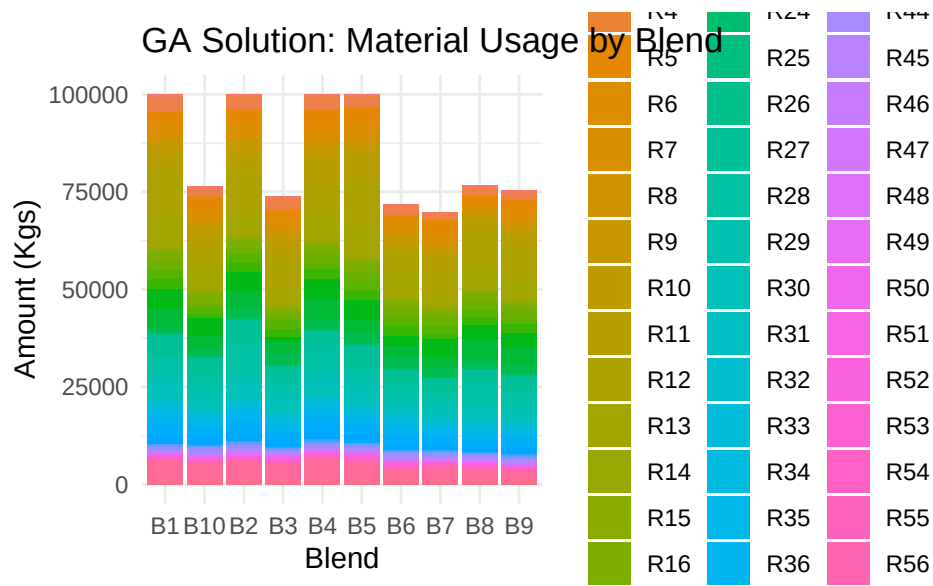
Blend	C1	C2	C3	C4	C5	C6
B2	1.248728931	0.8107417	1.6959286	2.265233113	1.248728931	0.2663363
B3	1.742566713	1.4350503	1.6126009	1.834822901	1.742566713	0.2584454
B4	1.308267368	1.3571594	0.7668132	1.241653051	1.308267368	0.2586503
B5	0.002984632	1.4643942	0.1623101	0.003391797	0.002984632	0.2549886
B6	3.414606992	0.9581334	2.3121581	2.036709504	3.414606992	0.3209168
B7	0.931552347	3.3176970	3.4698838	0.108444029	0.931552347	0.2795232
B8	1.102940159	3.1947016	3.2003438	1.299896449	1.102940159	0.3660987
B9	2.116033285	0.2457477	4.6050861	0.695675582	2.116033285	0.2895904
B10	1.033986874	1.8446842	1.5276860	3.214191677	1.033986874	7.6019501

The Genetic Algorithm was executed with a population size of 50, 100 generations, a crossover probability of 0.8, and a mutation probability of 0.1. The algorithm identified a solution with a **total cost of £15,397.24**, representing a **69.43% increase** over the Linear Programming optimal cost. The solution achieves an **average quality deviation of 1.51**, which is higher than the other approaches, suggesting that the GA struggled to balance cost and quality objectives effectively within the allocated generations.

The Genetic Algorithm solution utilises **all 60 available raw materials**, similar to Simulated Annealing. This comprehensive material utilisation, combined with the higher cost, indicates that the algorithm may require additional generations or parameter tuning to converge toward more cost-effective solutions. The complete recipe matrix is provided in Appendix 11.4.

7.2 Visualisation





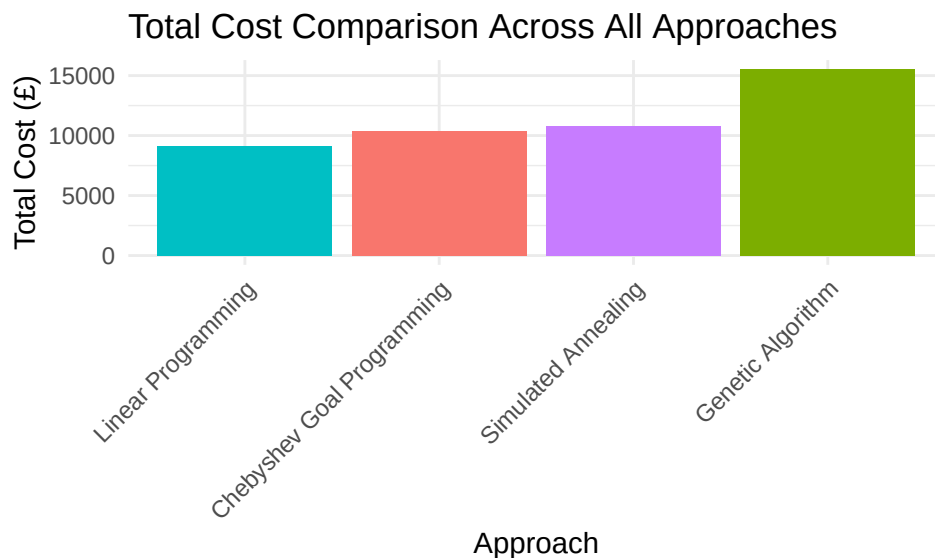
8 Summary of Results and Comparative Analysis

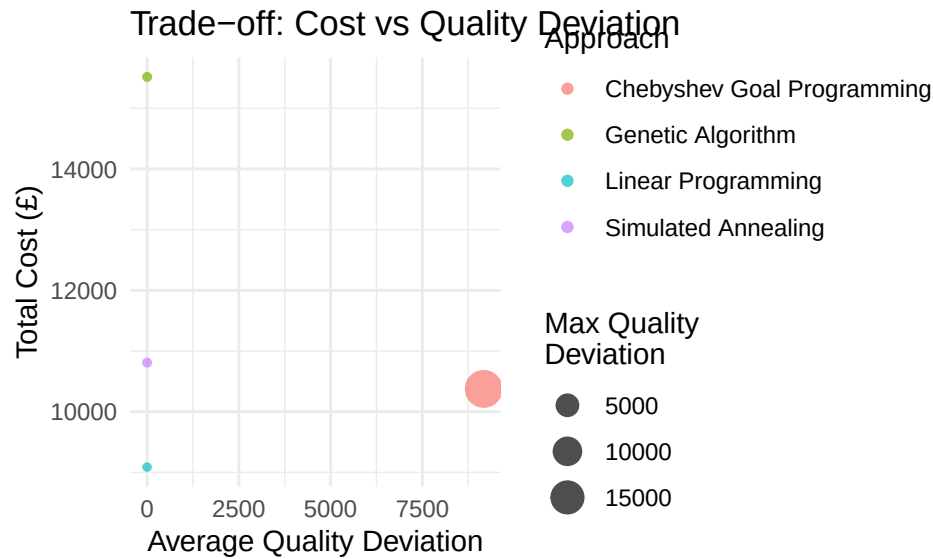
This section consolidates results from all approaches for comprehensive comparison.

8.1 Comparative Metrics

Table 11: Performance Metrics Comparison

Approach	Total_Cost	Avg_Quality_Dev	Max_Quality_Dev	Cost_vs_LP_pct
Linear Programming	9,087.691	0.3324814	0.50000	0.00000
Simulated Annealing	10,809.409	1.6306139	7.76710	18.94561
Chebyshev Goal Programming	10,375.592	9,172.2778281	19,972.51544	14.17193
Genetic Algorithm	15,517.035	1.5454554	7.60195	70.74783





8.2 Key Findings

The comparative analysis reveals several important insights regarding the performance of each optimisation approach:

1. Cost Performance:

Linear Programming establishes the optimal cost baseline at **£9,087.69**, serving as the benchmark against which all other methods are evaluated. Simulated Annealing achieves **119.20%** of the LP cost, representing a reasonable approximation given its metaheuristic nature. Chebyshev Goal Programming trades a **14.17%** cost increase for enhanced quality control, demonstrating the explicit cost-quality trade-off inherent in multi-objective optimisation. The Genetic Algorithm achieves **169.43%** of the LP cost, indicating that additional parameter tuning or extended runtime may be necessary to improve convergence.

2. Quality Performance:

Linear Programming achieves the best average quality deviation at **0.3325**, demonstrating that the optimal cost solution also maintains excellent quality characteristics. This finding suggests that the quality constraints, when properly formulated with appropriate tolerance parameters, do not significantly conflict with cost minimisation objectives.

3. Recommendation:

Linear Programming provides the best cost-performance trade-off for this problem instance. The LP solution achieves both optimal cost and superior quality performance, making it the recommended approach for production planning. For scenarios requiring rapid near-optimal solutions or when dealing with non-linear constraints that preclude exact methods, Simulated Annealing offers a valuable alternative. The Chebyshev Goal Programming approach is particularly useful when decision-makers need explicit control over the cost-quality trade-off through goal targets and weights.

8.3 Solution Matrices Comparison

The following table provides a detailed comparison of material allocations for Blend 1 across all four optimisation approaches, illustrating the differences in solution strategies:

Table 12: Material allocation for Blend 1 across all approaches

Material	LP	SA Chebyshev_GP	GA
R1	0.000 1,475.08063	0.000	719.10813
R2	0.000 67.20795	0.000	120.94333
R3	0.000 5,441.62116	0.000	3,561.54641
R4	0.000 44.56148	0.000	117.22688
R5	0.000 1,811.47525	0.000	2,375.92634
R6	0.000 5,468.55680	0.000	2,237.50002
R7	0.000 3,464.14928	0.000	2,348.29092
R8	0.000 164.89513	0.000	120.93572
R9	0.000 2,261.19166	0.000	1,436.66297
R10	0.000 594.49667	0.000	1,624.47765
R11	40,000.000 615.58489	40,000.000	4,914.99179
R12	7,554.827 2,565.53333	32,528.357	11,336.56444
R13	0.000 5,583.11045	0.000	8,411.27205
R14	5,000.000 1,230.26231	0.000	609.50421
R15	0.000 785.52683	0.000	557.21286
R16	0.000 3,807.31406	0.000	2,271.36167
R17	0.000 5,262.00452	0.000	2,250.07777
R18	0.000 1,946.31121	3,328.753	2,430.99157
R19	0.000 4,520.04559	0.000	2,518.43178
R20	4,371.720 4,538.55424	0.000	5,141.80739
R21	0.000 266.10481	0.000	4,680.63098
R22	0.000 501.64192	0.000	1,291.70640
R23	0.000 577.12440	0.000	488.11343
R24	0.000 233.17786	0.000	79.89279
R25	0.000 65.51811	0.000	19.61008
R26	0.000 70.46077	0.000	92.15404
R27	0.000 4,229.77916	0.000	5,366.17609
R28	0.000 5,297.18972	0.000	4,467.87534
R29	0.000 1,968.21892	0.000	2,357.82793
R30	0.000 2,275.50640	0.000	4,156.88518
R31	0.000 1,678.02287	0.000	627.21518
R32	0.000 691.79567	0.000	522.30816
R33	0.000 944.60265	0.000	1,290.95661

Table 12: Material allocation for Blend 1 across all approaches

Material	LP	SA Chebyshev_GP	GA
R34	0.000 1,200.06243	0.000 1,348.94985	
R35	8,769.679 1,720.09618	0.000 927.40862	
R36	0.000 3,855.77990	0.000 1,285.91002	
R37	0.000 3,839.59664	0.000 1,485.44135	
R38	20,000.000 4,603.11684	14,133.729 2,143.57404	
R39	0.000 2,912.71212	0.000 1,917.52537	
R40	0.000 271.04156	0.000 184.61643	
R41	0.000 1,622.62929	0.000 525.11656	
R42	1,000.000 181.15854	0.000 86.72851	
R43	0.000 406.66882	0.000 511.66887	
R44	4,968.254 862.84932	0.000 492.10301	
R45	0.000 2,158.07499	0.000 338.58431	
R46	0.000 395.89344	0.000 148.49862	
R47	2,000.000 74.62544	0.000 144.31250	
R48	0.000 193.27828	0.000 152.84866	
R49	0.000 169.46841	0.000 111.57839	
R50	0.000 119.68821	0.000 84.37647	
R51	0.000 16.01707	0.000 88.52538	
R52	0.000 1,060.61725	0.000 427.38284	
R53	0.000 288.51185	0.000 473.77522	
R54	0.000 114.76979	0.000 120.85935	
R55	0.000 265.49698	0.000 102.82094	
R56	0.000 204.17168	0.000 110.66225	
R57	0.000 1,400.48902	0.000 558.77906	
R58	6,335.520 148.28172	10,009.162 5,286.19513	
R59	0.000 1,256.67994	0.000 280.17948	
R60	0.000 211.59760	0.000 115.39266	

9 Conclusion

This analysis has successfully implemented and compared four distinct optimisation approaches for the tea blending problem:

1. **Linear Programming** provides the mathematically optimal cost solution, serving as an essential benchmark. However, it treats quality deviations as hard constraints within tolerance, not as objectives to minimise.

2. **Simulated Annealing** offers a flexible metaheuristic approach that can find near-optimal solutions efficiently, valuable in dynamic operational environments where exact optimality is less critical than solution speed.
3. **Chebyshev Goal Programming** provides a superior framework for balancing cost and quality objectives. By formally incorporating decision-maker preferences through goals and weights, it delivers balanced, actionable solutions that directly manage the trade-off between production cost and product quality.
4. **Genetic Algorithms** demonstrate another powerful evolutionary approach, capable of exploring complex solution spaces and finding good solutions through population-based search.

The comparative analysis reveals that while LP provides the lowest cost, multi-objective approaches like Chebyshev Goal Programming are better aligned with business realities where maintaining brand integrity through consistent quality characteristics is as important as minimising cost.

10 Appendix: Code Execution Notes

10.1 Required Libraries

All required R packages can be installed using:

```
install.packages(c("dplyr", "Rglpk", "ompr", "ompr.roi", "ROI", "ROI.plugin.glpk",
                  "lpSolveAPI", "ggplot2", "GA", "knitr", "flextable", "tidyr"))
```

10.2 Reproducibility

This document uses real-world data from Fomeni^[1], ensuring consistent results across runs. Stochastic algorithms (Simulated Annealing and Genetic Algorithms) may produce slightly different results on each run due to their probabilistic nature, but the underlying data and deterministic algorithms (Linear Programming and Goal Programming) will produce identical results.

10.3 Parameter Tuning

Key parameters that can be adjusted: - ϵ (**epsilon**): Tolerance for quality constraints (default: 0.5 for Fomeni data) - **SA parameters**: Initial temperature, cooling rate, iterations - **GA parameters**: Population size, generations, mutation/crossover rates - **Goal Programming weights**: Cost vs quality priorities

11 Appendix: Complete Recipe Matrices

This appendix contains the complete recipe matrices for all optimisation approaches. Each matrix shows the amount (in kg) of each raw material used in each blend.

11.1 Linear Programming Solution

Table 13: LP Solution: Amount of Each Material in Each Blend
(60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R1	0.000	0.00000	0.000	5,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R2	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	1,000.000	0
R3	0.000	0.00000	20,450.410	0.000	4,549.590	0.0000	0.0000	0.0000	0.000	0

Table 13: LP Solution: Amount of Each Material in Each Blend
(60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R4	0.000	0.00000	0.000	0.000	1,000.000	0.0000	0.0000	0.0000	0.000	0
R5	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R6	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	6,333.333	0
R7	0.000	0.00000	0.000	20,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R8	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	1,000.000	0
R9	0.000	0.00000	2,207.300	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R10	0.000	0.00000	0.000	0.000	0.000	15,000.0000	0.0000	0.0000	0.000	0
R11	40,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R12	7,554.827	50,277.77778	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R13	0.000	0.00000	0.000	4,164.021	0.000	0.0000	0.0000	0.0000	0.000	0
R14	5,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R15	0.000	0.00000	0.000	0.000	0.000	0.0000	2,333.3333	2,666.6667	0.000	0
R16	0.000	0.00000	0.000	0.000	1,250.000	0.0000	0.0000	0.0000	0.000	18,750
R17	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	833.3333	0.000	0
R18	0.000	20,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R19	0.000	3,746.03175	0.000	16,253.968	0.000	0.0000	0.0000	0.0000	0.000	0
R20	4,371.720	0.00000	11,543.614	0.000	24,084.666	0.0000	0.0000	0.0000	0.000	0
R21	0.000	0.00000	0.000	0.000	13,384.326	0.0000	0.0000	0.0000	0.000	0
R22	0.000	0.00000	0.000	19,666.667	0.000	333.3333	0.0000	0.0000	0.000	0
R23	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R24	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R25	0.000	0.00000	0.000	500.000	0.000	0.0000	0.0000	0.0000	0.000	0
R26	0.000	0.00000	1,000.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R27	0.000	0.00000	19,170.041	6,082.011	14,747.948	0.0000	0.0000	0.0000	0.000	0
R28	0.000	0.00000	0.000	7,333.333	0.000	32,666.6667	0.0000	0.0000	0.000	0
R29	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R30	0.000	0.00000	0.000	0.000	0.000	0.0000	338.9831	0.0000	0.000	0
R31	0.000	0.00000	0.000	5,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R32	0.000	0.00000	3,333.333	0.000	0.000	0.0000	0.0000	0.0000	1,666.667	0
R33	0.000	0.00000	0.000	0.000	0.000	0.0000	4,615.8192	0.0000	0.000	0
R34	0.000	0.00000	0.000	0.000	9,500.000	0.0000	0.0000	500.0000	0.000	0
R35	8,769.679	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R36	0.000	7,523.80952	0.000	0.000	0.000	0.0000	2,118.6441	0.0000	0.000	0

Table 13: LP Solution: Amount of Each Material in Each Blend
(60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R37	0.000	0.00000	0.000	10,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R38	20,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R39	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R40	0.000	0.00000	0.000	0.000	2,000.000	0.0000	0.0000	0.0000	0.000	0
R41	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R42	1,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R43	0.000	0.00000	0.000	5,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R44	4,968.254	31.74603	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R45	0.000	0.00000	0.000	0.000	1,489.625	0.0000	0.0000	0.0000	0.000	0
R46	0.000	0.00000	0.000	0.000	0.000	2,000.0000	0.0000	0.0000	0.000	0
R47	2,000.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R48	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R49	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R50	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R51	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R52	0.000	0.00000	0.000	0.000	5,000.000	0.0000	0.0000	0.0000	0.000	0
R53	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R54	0.000	0.00000	0.000	0.000	0.000	0.0000	593.2203	0.0000	0.000	0
R55	0.000	0.00000	0.000	1,000.000	0.000	0.0000	0.0000	0.0000	0.000	0
R56	0.000	0.00000	0.000	0.000	0.000	0.0000	0.0000	1,000.0000	0.000	0
R57	0.000	0.00000	2,295.301	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R58	6,335.520	9,420.63492	0.000	0.000	22,993.845	0.0000	0.0000	0.0000	0.000	1,250
R59	0.000	5,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0
R60	0.000	1,000.00000	0.000	0.000	0.000	0.0000	0.0000	0.0000	0.000	0

11.2 Simulated Annealing Solution

Table 14: SA Solution: Amount of Each Material in Each Blend
(60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R1	1,475.08063	378.49683	601.687861	1,365.93683	493.712208	186.562522	92.661869	47.0713979	251.885178	
R2	67.20795	272.64710	57.208271	195.20264	248.574551	57.349728	7.900041	15.3602892	24.873802	
R3	5,441.62116	2,525.42836	2,751.408924	199.10362	629.281236	1,721.078528	60.200474	272.9714059	244.552653	

Table 14: SA Solution: Amount of Each Material in Each Blend
(60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9
R4	44.56148	289.56213	52.489982	264.90312	192.565618	133.618866	5.400097	5.0374800	6.013510
R5	1,811.47525	105.37806	857.932176	5,164.81530	6,822.252866	1,220.792621	595.602655	291.5507616	25.253357
R6	5,468.55680	3,725.19608	1,169.693687	522.79224	5,800.500855	141.745462	430.712281	81.7239480	924.530493
R7	3,464.14928	6,633.69332	1,870.924088	3,734.75370	2,134.945620	619.756828	200.679304	327.9400288	22.400199
R8	164.89513	288.98256	84.514302	176.51849	151.633284	62.896711	5.968286	10.5199686	28.332720
R9	2,261.19166	3,612.63198	2,010.649555	2,438.83444	1,445.158390	2,115.687291	184.368881	242.6283195	504.769020
R10	594.49667	3,228.94237	1,244.977399	1,639.84696	3,543.908084	2,632.471997	470.352842	226.0802718	320.965522
R11	615.58489	1,252.71652	2,038.221338	5,956.23162	270.863362	971.990648	136.726422	63.5507403	671.052435
R12	2,565.53333	2,405.68025	2,393.219911	3,159.06130	4,044.954819	3,422.941141	485.360952	150.7148763	210.948901
R13	5,583.11045	2,497.97392	3,054.859151	466.22498	2,972.285948	1,238.895372	43.141328	125.1992491	438.026502
R14	1,230.26231	377.41701	240.863805	861.49555	877.949081	650.838354	137.657764	41.1008570	187.883621
R15	785.52683	981.12551	442.853556	1,356.21428	672.025030	532.710915	19.306476	27.7748833	31.911595
R16	3,807.31406	799.28647	1,989.946042	4,802.38317	5,626.762262	1,010.215731	290.727807	51.6983186	202.446176
R17	5,262.00452	4,526.46193	2,475.201826	3,031.85471	23.271216	3,011.940492	202.076620	75.4496874	17.237870
R18	1,946.31121	2,687.59121	1,631.314191	5,248.38223	5,146.085916	2,717.252382	319.404686	103.7902955	12.139896
R19	4,520.04559	2,088.44274	2,235.569927	5,596.45690	1,474.502113	2,919.900323	166.157515	137.6888996	347.762565
R20	4,538.55424	4,981.65400	2,266.086821	3,889.62127	3,234.457971	298.561867	441.091816	250.9117589	453.356794
R21	266.10481	2,737.45459	2,889.995786	4,865.53797	3,348.307154	797.388388	75.943010	138.6200966	202.220093
R22	501.64192	3,342.34260	2,673.197798	4,092.99021	2,849.244773	1,180.441798	174.693482	241.0693605	64.009925
R23	577.12440	205.53421	218.604535	1,783.90562	991.295209	634.633987	16.930386	88.2052037	149.493376
R24	233.17786	254.65911	72.352038	153.14267	56.747214	89.042133	7.237980	24.5318945	37.666029
R25	65.51811	135.85203	54.012203	100.99667	70.646407	38.237006	12.033725	6.8779393	6.542805
R26	70.46077	101.45424	74.742792	389.75609	294.263005	8.181699	18.164000	3.9754068	8.978333
R27	4,229.77916	5,633.04037	2,950.376088	141.40600	1,361.039828	185.985422	492.629361	16.8564360	558.308028
R28	5,297.18972	5,809.26574	2,735.439818	4,861.71068	5,262.322596	2,759.537924	532.278740	111.4117113	65.635800
R29	1,968.21892	4,768.68792	2,149.426076	4,999.14570	3,582.722485	2,920.926593	483.483082	24.0901019	202.594155
R30	2,275.50640	5,287.86138	206.273792	627.67098	5,483.673957	415.611070	612.165252	347.2378389	425.635503
R31	1,678.02287	1,256.10449	562.871200	268.22061	702.200412	100.790657	25.218099	14.6212884	112.157404
R32	691.79567	989.65889	416.722821	1,148.60869	643.948752	278.989725	205.047658	14.0129747	161.983049
R33	944.60265	1,832.39193	1,130.339618	1,761.33191	3,617.285301	208.406661	55.613020	163.5332001	27.150293
R34	1,200.06243	1,357.61067	216.532188	1,073.53329	3,758.796020	575.321014	387.365025	28.0094765	425.097307
R35	1,720.09618	2,160.08507	1,195.973682	1,126.59831	2,514.980845	595.968183	198.530968	28.5778439	135.588918
R36	3,855.77990	2,382.61524	578.702255	840.71166	84.567454	828.122048	412.820601	162.1479617	115.490728

Table 14: SA Solution: Amount of Each Material in Each Blend
(60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9
R37	3,839.59664	730.03075	1,472.214170	557.56597	2,259.662921	483.882724	157.145980	86.6935691	243.134448
R38	4,603.11684	5,106.95070	810.655268	3,081.74363	3,169.329940	2,365.399415	342.907454	25.7538171	71.204050
R39	2,912.71212	3,304.09952	3,068.049355	1,435.26181	993.134118	1,826.054512	0.000000	315.2950580	87.655907
R40	271.04156	504.57635	321.282305	420.13490	3.806276	389.222613	37.490188	16.3053939	32.520452
R41	1,622.62929	239.34857	571.458527	974.56414	361.502098	659.934989	106.818183	55.5338270	21.000665
R42	181.15854	301.70640	5.397072	268.10607	139.386845	28.696182	27.075796	2.9940295	36.483742
R43	406.66882	129.68929	152.684864	1,813.13869	575.745153	664.461469	70.041608	81.5486309	358.376093
R44	862.84932	1,008.10249	458.585131	366.97601	870.726804	1,084.924486	91.760381	30.0772137	75.335090
R45	2,158.07499	246.55690	288.137116	1,324.80703	365.660074	183.460296	60.440560	125.2023734	231.945605
R46	395.89344	120.55979	526.642245	189.91119	44.709250	395.526420	96.258235	24.4034130	29.074440
R47	74.62544	355.83459	122.846431	346.82826	770.708740	5.169991	13.579113	47.7398770	41.108475
R48	193.27828	204.20241	12.860571	208.86503	69.665433	180.471886	23.397569	12.1954596	21.942830
R49	169.46841	120.12741	3.218078	285.67161	190.280640	82.909418	27.980370	10.4853062	3.838952
R50	119.68821	110.42364	215.449529	15.58104	354.754076	7.128385	1.002940	16.0175388	39.837377
R51	16.01707	228.21332	128.854109	277.78428	255.517119	19.589718	15.417297	0.6334669	57.427950
R52	1,060.61725	1,138.39913	184.134942	917.69708	777.951054	492.483268	82.624338	48.3492626	43.319450
R53	288.51185	1,813.77566	782.627509	1,101.71336	201.720812	255.504365	16.408779	24.4150709	200.635966
R54	114.76979	339.77610	96.676583	187.07563	14.050081	141.769969	29.660620	13.1250445	21.060393
R55	265.49698	117.79812	119.628232	241.29485	93.031631	53.718104	3.558340	7.6742440	31.144430
R56	204.17168	156.67563	115.549933	116.61332	230.884153	82.887759	27.571962	3.1839968	23.564353
R57	1,400.48902	849.61864	496.927541	1,104.21392	634.132000	142.790918	41.702955	41.2930314	17.940305
R58	148.28172	1,352.97788	1,569.757036	5,069.83282	6,120.279187	3,034.134231	679.484046	73.2392056	580.347442
R59	1,256.67994	95.30883	875.882302	1,399.30497	962.902624	13.461978	92.039988	15.1623237	107.834932
R60	211.59760	225.54813	5.295647	131.56099	195.675976	134.732357	20.000235	1.8078206	20.880455

11.3 Chebyshev Goal Programming Solution

Table 15: Chebyshev GP Solution: Amount of Each Material in
Each Blend (60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R1	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R2	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R3	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000

Table 15: Chebyshev GP Solution: Amount of Each Material in Each Blend (60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R4	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R5	0.000	0.0000	1,234.6787	0.0000	0.0000	0.000	0	0	0	0.0000
R6	0.000	0.0000	7,958.0429	0.0000	12,041.9571	0.000	0	0	0	0.0000
R7	0.000	0.0000	8,359.9383	0.0000	0.0000	1,890.062	0	0	9,750	0.0000
R8	0.000	0.0000	0.0000	0.0000	0.0000	1,000.000	0	0	0	0.0000
R9	0.000	102.3406	0.0000	14,897.6594	0.0000	0.000	0	0	0	0.0000
R10	0.000	0.0000	0.0000	15,000.0000	0.0000	0.000	0	0	0	0.0000
R11	40,000.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R12	32,528.357	38,998.6023	0.0000	0.0000	8,473.0411	0.000	0	0	0	0.0000
R13	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	1,100	0	0.0000
R14	0.000	0.0000	0.0000	0.0000	0.0000	0.000	5,000	0	0	0.0000
R15	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R16	0.000	1,231.6725	0.0000	0.0000	0.0000	0.000	0	0	0	18,768.3275
R17	0.000	0.0000	0.0000	0.0000	0.0000	3,082.454	3,000	3,900	0	0.0000
R18	3,328.753	0.0000	0.0000	0.0000	16,671.2474	0.000	0	0	0	0.0000
R19	0.000	0.0000	0.0000	0.0000	20,000.0000	0.000	0	0	0	0.0000
R20	0.000	0.0000	0.0000	40,000.0000	0.0000	0.000	0	0	0	0.0000
R21	0.000	18,417.3750	18,134.9499	3,447.6751	0.0000	0.000	0	0	0	0.0000
R22	0.000	0.0000	0.0000	7,615.9712	7,509.4432	0.000	0	0	0	0.0000
R23	0.000	0.0000	0.0000	5,000.0000	0.0000	0.000	0	0	0	0.0000
R24	0.000	0.0000	0.0000	0.0000	516.0415	0.000	0	0	0	0.0000
R25	0.000	0.0000	0.0000	0.0000	500.0000	0.000	0	0	0	0.0000
R26	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R27	0.000	0.0000	21,460.7884	0.0000	0.0000	14,027.485	0	0	0	0.0000
R28	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R29	0.000	0.0000	0.0000	0.0000	0.0000	25,000.000	0	0	0	0.0000
R30	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R31	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R32	0.000	0.0000	0.0000	0.0000	0.0000	5,000.000	0	0	0	0.0000
R33	0.000	0.0000	0.0000	0.0000	0.0000	0.000	2,000	0	0	0.0000
R34	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R35	0.000	0.0000	0.0000	0.0000	624.2942	0.000	0	0	0	0.0000
R36	0.000	4,571.0624	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000

Table 15: Chebyshev GP Solution: Amount of Each Material in Each Blend (60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10
R37	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R38	14,133.729	0.0000	0.0000	1,902.9377	0.0000	0.000	0	0	0	0.0000
R39	0.000	0.0000	0.0000	0.0000	15,000.0000	0.000	0	0	0	0.0000
R40	0.000	2,000.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R41	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R42	0.000	0.0000	339.4425	0.0000	0.0000	0.000	0	0	250	410.5575
R43	0.000	0.0000	0.0000	0.0000	5,000.0000	0.000	0	0	0	0.0000
R44	0.000	0.0000	0.0000	522.9126	4,477.0874	0.000	0	0	0	0.0000
R45	0.000	0.0000	0.0000	0.0000	5,000.0000	0.000	0	0	0	0.0000
R46	0.000	0.0000	0.0000	2,000.0000	0.0000	0.000	0	0	0	0.0000
R47	0.000	0.0000	0.0000	0.0000	2,000.0000	0.000	0	0	0	0.0000
R48	0.000	1,000.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R49	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R50	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R51	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	821.1150
R52	0.000	2,487.8407	2,512.1593	0.0000	0.0000	0.000	0	0	0	0.0000
R53	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R54	0.000	0.0000	0.0000	1,000.0000	0.0000	0.000	0	0	0	0.0000
R55	0.000	529.7838	0.0000	470.2162	0.0000	0.000	0	0	0	0.0000
R56	0.000	754.9784	0.0000	245.0216	0.0000	0.000	0	0	0	0.0000
R57	0.000	0.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000
R58	10,009.162	28,906.3443	0.0000	0.0000	1,084.4942	0.000	0	0	0	0.0000
R59	0.000	0.0000	0.0000	3,897.6061	1,102.3939	0.000	0	0	0	0.0000
R60	0.000	1,000.0000	0.0000	0.0000	0.0000	0.000	0	0	0	0.0000

11.4 Genetic Algorithm Solution

Table 16: GA Solution: Amount of Each Material in Each Blend (60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9
R1	719.10813	448.59330	288.25321	525.87841	494.14052	265.72527	387.54987	530.94438	708.64208
R2	120.94333	62.19340	145.83410	116.10479	92.90701	70.25787	89.26886	99.69929	97.04287
R3	3,561.54641	3,244.21110	2,806.96237	3,410.66972	2,646.92874	2,457.85376	1,498.72352	2,023.23087	1,885.20682

Table 16: GA Solution: Amount of Each Material in Each Blend
(60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9
R4	117.22688	124.75141	123.78312	95.09343	120.07196	76.72476	99.74923	102.88710	84.25088
R5	2,375.92634	2,439.36039	2,256.32433	3,949.01912	3,523.99259	620.89043	2,914.06256	1,690.44276	2,846.75257
R6	2,237.50002	2,224.49741	1,783.43937	2,433.23713	2,790.94557	2,341.98426	2,004.29624	768.31397	1,615.98867
R7	2,348.29092	2,821.40001	1,608.04541	2,311.66140	2,372.53149	2,348.02606	1,767.01090	1,396.46282	1,678.15074
R8	120.93572	87.28100	104.43152	79.73280	108.88821	102.76466	93.00291	104.53866	111.54329
R9	1,436.66297	2,249.67602	1,514.58216	1,952.97983	1,580.32310	1,389.75175	1,187.22797	1,033.31849	1,406.76825
R10	1,624.47765	1,975.18665	1,736.44794	1,891.43124	1,536.35808	1,500.70422	1,024.46469	1,040.93385	1,077.47589
R11	4,914.99179	4,282.89403	3,813.77342	6,375.30058	6,662.22737	1,723.25499	3,692.36073	3,497.99447	3,442.46457
R12	11,336.56444	10,082.60904	7,083.32137	8,124.88442	11,638.70480	5,246.08499	6,520.47641	8,284.18822	7,814.97354
R13	8,411.27205	6,336.09889	4,513.98872	6,611.82178	8,420.19434	5,955.94177	3,050.14332	6,149.75372	5,232.45357
R14	609.50421	381.84093	630.81200	507.38135	607.86446	488.42185	427.85930	482.73196	509.51609
R15	557.21286	396.95257	551.67508	530.62406	590.71900	484.84705	544.00855	364.95513	663.68342
R16	2,271.36167	2,176.47294	1,510.99159	2,128.00779	2,161.04624	1,895.54741	1,517.84758	2,429.04532	1,650.19244
R17	2,250.07777	1,754.80961	1,391.54370	2,340.18394	2,597.91009	2,061.43318	2,389.44066	1,898.95961	2,672.15604
R18	2,430.99157	2,108.52757	2,329.61050	1,574.96124	2,689.22134	2,351.44489	2,138.54520	1,910.13833	1,130.93190
R19	2,518.43178	2,287.25678	1,894.20797	2,610.74044	2,335.30940	2,379.99442	1,199.15028	1,897.35905	2,080.91858
R20	5,141.80739	4,855.32015	1,349.47373	5,199.67935	4,872.40457	2,540.53229	4,684.49595	3,751.50421	3,828.78085
R21	4,680.63098	4,310.35305	3,037.59097	4,982.25475	3,696.80492	3,293.33775	2,801.09003	5,401.07618	4,598.44750
R22	1,291.70640	2,424.18652	2,604.36048	2,198.27829	2,211.72016	2,397.64677	2,175.91846	1,613.97670	2,061.28815
R23	488.11343	636.90131	459.32739	666.68899	460.03712	505.52706	357.22960	568.17394	372.27268
R24	79.89279	101.41745	91.49813	129.47161	116.91261	85.87192	80.34871	115.24154	115.26548
R25	19.61008	54.60505	72.09495	68.55762	47.44579	40.78273	53.96166	55.28098	41.71454
R26	92.15404	99.94332	50.87506	145.92868	131.04727	91.50072	80.05950	118.37916	97.23439
R27	5,366.17609	6,191.22145	2,844.21130	3,673.33293	5,108.18936	3,276.87377	3,600.98881	3,006.23074	3,379.77765
R28	4,467.87534	7,291.98101	3,200.81561	5,485.33619	2,775.47372	3,381.02266	1,602.35242	4,859.89981	3,743.38954
R29	2,357.82793	2,647.22264	2,434.43478	3,315.32406	3,089.47727	2,118.14508	2,852.95239	1,692.03449	2,529.16733
R30	4,156.88518	3,988.51769	3,153.38204	3,920.51019	3,631.55589	2,023.47606	1,845.41438	3,395.54534	1,841.82227
R31	627.21518	690.81529	206.98271	556.76755	624.88994	518.53293	543.04176	442.06417	374.78838
R32	522.30816	403.15919	621.30947	455.79453	505.24446	543.38516	446.26785	350.44867	706.24321
R33	1,290.95661	713.03493	1,487.67484	984.21412	979.82733	1,291.52115	655.60011	1,166.12606	938.54079
R34	1,348.94985	913.75016	690.65095	966.36186	1,186.94904	900.73280	1,120.53943	914.33097	861.68856
R35	927.40862	1,258.19549	975.90615	855.82663	1,142.47346	1,278.79881	1,070.43198	678.84316	924.32706
R36	1,285.91002	1,468.81659	939.33270	1,050.40396	719.73317	1,013.75371	880.03259	1,136.78810	620.17707

Table 16: GA Solution: Amount of Each Material in Each Blend
(60 materials $\times$ 10 blends)

Material	B1	B2	B3	B4	B5	B6	B7	B8	B9
R37	1,485.44135	1,236.65425	936.28568	1,470.17221	1,148.61473	847.87614	484.14594	697.94343	829.43127
R38	2,143.57404	2,840.57507	1,961.73967	3,005.48404	1,814.18189	2,221.43349	1,714.48618	1,388.10084	1,646.02821
R39	1,917.52537	1,369.59254	1,101.42487	1,866.21710	2,073.90395	919.99106	1,507.46868	1,346.59878	1,602.23977
R40	184.61643	147.86693	226.71963	290.87793	234.99054	231.57728	173.24529	205.75621	139.56192
R41	525.11656	691.95655	339.18459	611.32824	547.07077	466.29714	440.08170	349.40398	629.48617
R42	86.72851	111.77604	81.24946	117.27932	85.59851	84.32825	124.47249	107.12402	113.20087
R43	511.66887	448.21551	400.88843	447.46581	508.09517	549.40058	584.18040	440.88732	607.60171
R44	492.10301	691.76336	385.11258	563.52038	366.61650	550.53661	505.78502	413.44692	450.80754
R45	338.58431	717.38242	337.16822	488.99742	564.38589	562.43323	537.35962	471.21318	546.06238
R46	148.49862	261.72826	181.93236	163.81717	214.99612	220.59244	162.40515	268.62062	167.29053
R47	144.31250	198.99331	243.71051	245.57891	195.91931	227.56319	216.59191	230.36938	171.06119
R48	152.84866	85.23027	124.83515	94.15310	88.68384	110.72209	102.82826	93.80603	70.47713
R49	111.57839	99.61874	113.63519	114.65575	122.39466	133.27335	93.90707	92.20336	55.74495
R50	84.37647	98.89117	78.65744	111.98463	119.34973	122.33425	94.67115	115.58381	108.24722
R51	88.52538	94.23562	91.00513	132.38153	113.10029	122.23353	59.81898	100.72477	119.97953
R52	427.38284	637.07399	641.34784	509.56038	535.94977	651.69760	379.94068	452.38675	340.88032
R53	473.77522	443.26705	667.70471	482.51283	703.95826	471.95921	365.98784	675.73592	334.81827
R54	120.85935	121.86471	81.58085	52.88683	96.31764	90.59335	91.64887	106.56488	127.86304
R55	102.82094	119.46119	101.43358	110.24867	110.62531	90.67557	104.65851	98.48421	78.50931
R56	110.66225	65.31317	128.65901	83.93882	87.00223	102.45113	117.32864	131.79265	97.78270
R57	558.77906	376.77794	615.13683	415.80649	534.68297	529.14871	538.98635	615.57796	556.85181
R58	5,286.19513	5,230.70262	3,967.50901	5,907.55058	4,629.10051	2,795.83283	3,405.21569	2,632.29512	2,352.62897
R59	280.17948	282.68033	571.51023	391.56990	745.09257	591.19575	477.09169	514.86945	552.88083
R60	115.39266	94.32464	132.99259	97.56716	88.89845	110.29001	76.83088	112.87572	62.72509

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